UNIVERSITY RESEARCH AND DEVELOPMENT AGENDA (2021-2024)

Thematic Areas

A. COVID 19 Pandemic

1. Online Flexible Learning System
2. Gamification of learning in education
3. Online digital transactions for food commodities
4. Fabrication of Small Multi-task Agri-Entrep Machines
5. Nutraceuticals and Enhanced Foods
6. Impact Evaluation of COVID on SMEs
7. Development on PPEs from Local Materials
8. Flexible Work Arrangements and Social Distancing
9. Automation and Digital Technology
10. Characterization and Management of Infectious Wastes
11. Urban Agriculture and Behavior of People Towards Environment
12. Food Safety
13. Food Innovation
14. Creative Industry and Heritage
15. COVID Social Science and Policy Studies

B. Crops

1. Crop Production System Approach
2. Post-harvest, primary processing and Product Development

C. Livestock

1. Breed development and genetic improvement
2. Reproductive bio techniques for priority livestock species
3. Nutrition, feeds and feeding system
4. Conservation and improvement of native animals
5. Food quality and safety assurance
6. Disease control and anti-microbial resistance management
7. Production and management decision support systems
8. 1Product development and processing

D. Aquatic

1. Application of genomics in the study of diseases of aquatic species and fish resistance to climate change; molecular phylogenetics; population genetics
2. Physiological and ecological studies of aquatic species
3. New species for culture
4. Culture systems (brood stock management, hatchery, nursery, grow-out)
5. Fish health, disease diagnostics and disease management
6. Nutrition, feeds and feeding system
7. Postharvest handling, processing and new product development
8. Production and management decision support systems

E. Forestry

1. Physiological and ecological studies
2. Varietal improvement of priority agroforestry/non-timber and forestry species
3. Sustainable production, management and processing of priority agroforestry/non-timber and forestry species
4. Decision support systems

F. Natural Resources and Environment

1. Conservation, sustainable utilization, and management of biodiversity in terrestrial, freshwater and marine ecosystems
2. Sustainable inland and coastal watershed management and utilization
3. Strategies/decision management tools for climate change resilient environment
4. Resource and ecosystems assessment and monitoring
5. Habitat management for fishery and ecosystem sustainability
6. Marine environmental management (coastal integrity/erosion and eutrophication)
7. Innovative management systems for unique landscapes and ecosystems
8. Ecotourism management

G. **Technology Transfer**

1. Innovative and improvement of traditional extension modalities for the efficient transfer of technologies to end-users
2. Upscaling and out scaling of agricultural technology transfer and commercialization through various modalities

H. **Socio-Economics, Policy Research, and Governance**

1. Agriculture and resource economics studies such as but not limited to market research, agrarian/asset reform, environmental valuation, economies of scale/collective farming
2. Supply/value chain development of selected AANR industries and other agribusiness-related initiative
3. Policy research and advocacy on natural resources/environment-related issues, agricultural trade, supply chain/value chain related issues, standards and regulations, compliance to standards across the value chain, and R&D governance and other cross-cutting policy concerns
4. Integrating gender and development (GAD) in AANR programs and projects.
5. Software technologies towards better foresight and strategic insights.
6. Applied rural sociology studies covering but not limited to characterization and understanding rural communities; demographic and intergenerational changes of AANR stakeholders (e.g. farmers, fishers, indigenous people); social practices and behavior and their implications to technology adoption; role of social institutions and institutional arrangements.

I. **Human Capital Systems**

1. Systems approach to knowledge management and cross R&D collaborations

J. **Health**

1. Responsive Health Systems
2. Research to enhance and extend healthy lives
3. Holistic approaches to health and wellness
4. Health resiliency
5. Research in equity and health

REFERENCES:
DOST-PCAARRD Updated Harmonized National Research and Development Agenda 2022-2028
Partido State University Research and Extension Direction and Focus for 2020-2022
PCHRD National Unified Research Agenda (2017-2022)

LUISA M. LANCISO
Director, Research and Development

PATRICIA M. CANDELARIA
Vice President for Research, Extension & Knowledge Management